SXM GO

Gamified Island Exploration

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SPRINT 3

PRESENTATION

TEAM MEMBERS

- Oliver Skoczylas (Dev 1) – Core Auth & User Management
- Kaiden Vandegrift (Dev 2) – Map, Location & Check-In
- Taylor Knipe (Dev 3) – Gamification Engine
- Paul Han (Dev 4) – Social Features & Leaderboards

https://sxmgo.vercel.app/
https://github.com/OliverSkoczylas/SXM-GO

PRIORITIZED SPRINT 3 GOALS

P1

Itinerary Builder UI

Full itinerary creation and editing UI (3–20 stops); pre-made routes: Foodie, Beach Hopper, Cultural Explorer; share via unique link + QR code.

Owner: Dev 4

P1

Activity Route Tracking

GPS start/stop route recording; route displayed on map; activity statistics: distance, duration, pace, elevation.

Owner: Dev 3

P1

Friend System & Groups

Friend search, requests, and profiles; group competition with invite codes and group leaderboard.

Owner: Dev 1 & Dev 4

P2

Full Push Notification Pipeline

End-to-end FCM/APNs delivery for badge, challenge, and leaderboard events; <30s delivery SLA.

Owner: Dev 4

P2

GDPR/CCPA Compliance + Profile UI

Data export (JSON), consent controls, deletion; profile update screen (name, photo, preferences).

Owner: Dev 1

P3

Map Polish & Infra

Route overlay on map; location detail view polish; clustering for 50+ pins; Supabase paid tier upgrade; WCAG 2.1 AA accessibility audit.

Owner: Dev 2 / All

PROJECT BACKLOG · SPRINT BACKLOG

PROJECT BACKLOG (Selected Items)

FR-001–013	User Auth & Profiles	DONE
FR-014–026	Map, Locations & Check-In	DONE
FR-027–035	GPS Anti-Fraud & Security	DONE
FR-036–043	Points, Badges & Challenges	DONE
FR-072–075	Leaderboard System	DONE
FR-044–058	Itinerary System	DONE
FR-084–096	Activity / Route Tracking	DONE
FR-097–103	Social Features & Friends	DONE
TR-027–035	GDPR / Compliance	DONE

SPRINT 3 BACKLOG

Itinerary Builder UI (create, edit, share)	Dev 4
Pre-made itineraries (Foodie, Beach, Cultural)	Dev 4
Full push notification delivery pipeline	Dev 4
Group competition & invite code system	Dev 4
Friend system UI (search, requests, profiles)	Dev 1 & 4
Friend system backend (send/accept/decline)	Dev 1
GPS activity route tracking (start/stop)	Dev 2
Activity statistics screen	Dev 2
Streak bonus & badge tier UI	Dev 3
Route visualization overlay on map	Dev 2
Location detail view polish	Dev 2
Profile update UI & GDPR compliance	Dev 1
Supabase paid tier + clustering	Dev 2
Accessibility audit (WCAG 2.1 AA)	All

🔒 CYBERSECURITY ASSESSMENT

Authentication Security

COMPLETE

- Passwords bcrypt hashed
- JWTs in Keychain/Keystore
- Sessions expire 30d; rate limiting 5/min on login
- Google & Apple OAuth live

Data Privacy & GDPR

IN PROGRESS

- HTTPS/TLS 1.3 on all endpoints
- GDPR data export endpoint (Sprint 3)
- CCPA consent controls (Sprint 3)
- User deletion operational

GPS Anti-Fraud

COMPLETE

- Impossible-travel-speed detection live
- Mock location flag detection active
- QR code fallback for poor GPS zones
- Manual review queue for flagged check-ins

API Security

IN PROGRESS

- API keys server-side via proxy
- Input validation on all endpoints
- SQL injection protection via Supabase ORM
- OWASP Top 10 audit planned Sprint 3

SPONSOR FEEDBACK

CURRENT STATE

- Sponsor was very satisfied with the current product
- Faculty advisor liked the current feature set on the user side
- Only questions were about adding business functionality

LOOKING FORWARD

- Next team will be responsible for adding the business portal
- They will also be in charge of final deployment for the UA team

LESSONS LEARNED

1. OAuth Config Takes Longer Than Estimated

Provider edge cases (Apple on Android) require extra time. Budget 1.5× the estimate for OAuth work.

2. Seed Data Quality Matters Early

Placeholder pins slowed Dev 2. Real coordinates should be seeded before frontend pin work starts.

3. Security Must Be Sprint 1 Scope

TLS rate limiting on all routes should have been in Sprint 1. Retrofitting security costs more than building it in.

4. Integration Tests Before Demo

Check-in → points → badge pipeline had edge cases found late. Integration tests before sprint review are mandatory.

5. Supabase Realtime Tier Planning

Free tier connection limits hit during load test. Upgrading to paid tier is required.

TEAM CONTRIBUTIONS

Developer 1 – Oliver Skoczylas

Auth, Security & GDPR

Completing:

Profile update UI, GDPR data export + consent, friend system backend, security audit

Developer 2 – Kaiden Vandegrift

Map, Route & Polish

Completing:

Route overlay on map, location detail polish, map clustering, Supabase upgrade

Developer 3 – Taylor Knipe

Gamification & Activity

Completing:

Project Documentation
API Standardization and Docs
Test suite for shared components

Developer 4 – Paul Han

Social, Itineraries & Notifications

Completing:

Leaderboard UI Refinements
Itinerary builder UI Refinements
Itinerary sharing
Social sharing feature

LIVE DEMO

Sprint 2 Feature Walkthrough

01	02	03	04
Social OAuth (Google Login)	Offline Map & QR Check-In	Badge Unlock Notification	Weekly Leaderboard